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When Does SHAP Outperform LIME? Model- and Configuration-Dependent Results from a Paired Tabular Benchmark

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Abstract:

~~LIME and SHAP are the most widely used model-agnostic explainers for tabular models, yet controlled paired comparisons of the two remain scarce. This paper extends an earlier paired test to the full grid of a public benchmark: 75 cells that share model, seed and sample size, over five model families on the UCI Adult census-income dataset, with Holm-corrected paired tests and effect sizes for five endpoints. SHAP leads on stability (d_z = 3.00), fidelity (d_z = 4.82) and faithfulness gap (d_z = 2.63), and LIME gives sparser explanations. Two results depend on conditions that are usually left implicit. First, the latency ordering depends on the model family: SHAP's TreeExplainer was faster than LIME in every XGBoost cell and slower in every random-forest cell, and LIME was faster for non-tree models (median 53.3 ms vs. 694.6 ms). Second, LIME's near-zero stability on Adult is not a fixed property of the method: it is high on two further tabular datasets and, in a single-model probe, rises with the kernel width. The results support choosing the explainer according to the model architecture and the latency budget, and they show that a ranking obtained on one dataset and one configuration does not transfer by default. All artifacts are public.~~

Choosing a post-hoc explainer for a tabular model trades faithfulness, stability and computational cost, yet comparisons under matched conditions are scarce. This paper reports a paired benchmark of SHAP, LIME, Anchors and DiCE in which every explainer sees the same model, test instances and seed on one recorded host: five model families on the Adult dataset (confirmatory), six further datasets and a configuration sweep (exploratory). Cost is measured in time and in model evaluations, which do not depend on the hardware. On Adult, SHAP's fidelity exceeds that of each alternative in all 25 model-seed cells (d_z = 3.3 or more), but its stability advantage over LIME and Anchors is not significant; DiCE ranks last on every quality endpoint. On the six further datasets SHAP keeps the highest fidelity, while LIME is the more stable explainer, as it is on Adult with category-preserving noise or a wider kernel; SHAP's fidelity lead holds over every configuration tested. KernelSHAP's median wall time is 3 to 25 times LIME's, whereas TreeSHAP's cost follows tree depth: faster than LIME on XGBoost, and on the deeper random forest faster or slower by dataset. The choice of explainer therefore depends on the model, the configuration and the latency budget.

Keywords: *Benchmarking, Computational cost, Explainable artificial intelligence, Explanation evaluation, ~~LIME, SHAP~~Feature attribution*

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1. INTRODUCTION

A team that deploys a tabular classifier and must explain its predictions faces a concrete decision: which post-hoc explainer to use, for which model, under which latency budget. The candidates differ in the kind of explanation they return. LIME [1] and SHAP [2] ~~are the two most widely used model-agnostic post-hoc explainers for tabular models. A practitioner who must choose one of them needs to know~~return feature attributions, Anchors [3] returns a rule and DiCE [4] a counterfactual. They also differ in how ~~they differ in~~faithfully the ~~properties that matter for the deployment: whether an~~ explanation tracks the model, ~~whether~~how stable it is ~~stable, how many features it uses and how long it takes. Their theoretical differences are well known, but they do~~ under small changes of the input, and how much computation it needs [5–7]. Theory does not ~~by themselves~~ settle the choice across model families~~, sampling regimes and runtime constraints [3, 4].~~

~~The~~ and configurations [8, 9], and the empirical guidance is weaker than the popularity of the ~~two~~ methods suggests~~. Evaluation of explanations is spread over several~~: evaluations use different metric families, and ~~comparisons between~~ the ranking of explainers often ~~depend~~depends as much on the metric ~~chosen~~and the configuration as on the methods ~~themselves [5–7]. Work that uses several metrics shows that fidelity, stability, sparsity and cost carry complementary signals [8, 9]. Controlled comparisons of LIME and SHAP under matched conditions, where the two explainers see the same model, the same instances and the same random seed, remain scarce. Without that pairing, a difference between the methods cannot be separated from a difference between the models or the samples they were run on.~~ [10–12].

Two gaps follow. The first is *control*. Comparisons in which all explainers see the same model, the same instances, the same seed and the same host, so that a difference can be attributed to the explainer, are scarce: published SHAP–LIME comparisons are often case studies in a single domain [13, 14]. Cost needs the same control: it transfers to another deployment only when it is measured on a stated host, as item 2c of the REFORMS checklist for machine-learning-based science recommends [15], and when it is also counted in a unit that does not depend on that host, such as the number of model calls. The second gap is *configuration*. Results are usually reported for one configuration, although the hyperparameters that build LIME's neighborhood [16] or bound KernelSHAP's sampling [7] can change the outcome. Without control and without more than one configuration, a recommendation cannot state the conditions under which it holds.

This paper addresses the two gaps with three contributions. The first closes the gap in control, the second the gap in configuration, and the third states what the results support once both are closed:

1. **A controlled, paired benchmark of four explainers** (SHAP, LIME, Anchors and DiCE) on five model families, with every run on one recorded host, and cost reported as time on that host and as the number of model evaluations, a measure that does not depend on the hardware (Sections 4.1 and 4.2).

2. ~~This paper reports such a paired comparison. It reuses the execution cohort released with an earlier study [10], which introduced the measurement protocol and compared four explainer families. That study already contains a first paired test of SHAP against LIME, on the 45 matched cells of three model families (logistic regression, random forest and XGBoost): SHAP was higher on fidelity, stability and faithfulness gap, and LIME was cheaper and sparser. It announced the analysis on the full grid of 75 cells without reporting it, and it reported averages over conditions only. The present paper completes that contrast on the 75 cells of the five model families, with confidence intervals and effect sizes, and then asks the question that an average over conditions hides: *when does the answer depend on the model and on the configuration?*~~

Configuration-aware results: a sweep over the LIME kernel width and sample size and over the KernelSHAP budget and background, and controls that test whether the stability measurement itself is valid (Sections 4.3 and 4.4).

3. **Bounded recommendations**: every claim is labelled as confirmatory or exploratory and stated with the models, datasets and configurations under which it holds (Section 5).

The paper makes ~~three contributions:~~

~~1. **Paired inferential evidence on the full grid.** The SHAP–LIME difference, tested before on a 45-cell subset, is quantified on the 75 matched cells of the UCI Adult census-income dataset, over five model families, with Holm-corrected paired tests, confidence intervals and effect sizes for five endpoints (Section 4).~~

~~2. **Two conditional results.** The latency ordering of the two methods depends on the model family, and reverses between two tree ensembles (Section 4.4). LIME's near-zero stability on Adult depends on the kernel width and on the feature space, and is high on two further datasets (Sections 4.5 and 4.6).~~

~~3. **A deployment recommendation conditioned on the model architecture**, with an explicit statement of which claims generalize and which do not (Section 5).~~

~~The paper makes~~ ranking claims under a stated protocol. It does not claim that ~~either method~~an explanation is correct for a task, and it does not test whether ~~users~~people decide better with ~~either~~any explainer.

2. BACKGROUND AND RELATED WORK

~~2.1.~~ LIME

~~LIME approximates a black-box model f locally around an instance x by training~~ fits an interpretable surrogate g to the model f on perturbed samples around an instance x, weighted by a proximity kernel π_x, as in Eq. (1):

$$\xi(x) = \operatorname{argmin}_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g), \tag{1}$$

where ~~$\xi(x)$ is the explanation returned for x,~~ G is the class of ~~interpretable~~ surrogates (sparse linear models here), $\mathcal{L}$ measures how poorly g reproduces f on ~~perturbed samples, each sample~~the weighted ~~by its proximity π_x to x,~~samples and $\Omega(g)$ penalizes surrogate complexity [1]. ~~LIME's behavior is shaped by the hyperparameters that build the neighborhood: the~~The kernel width, the number of ~~perturbed~~ samples and the feature budget~~. These controls yield compact explanations~~ shape the neighborhood, and ~~low latency, but they introduce~~ the sampling makes the explanation stochastic~~ variation, because local neighborhoods are sampled rather than exhaustively enumerated.~~

~~2.2~~. SHAP

~~SHAP explains a prediction through~~ assigns additive ~~feature~~ attributions $f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i$~~, where ϕ_0 is the model's expected output over a background sample, M is the number of features and ϕ_i is the Shapley-value contribution of feature i. SHAP values are the unique additive attributions~~ that satisfy local accuracy, missingness and consistency~~ [2].~~, where ϕ_0 is the expected output over a background sample and M the number of features [2].~~ KernelSHAP estimates them ~~with~~by a weighted regression over sampled feature coalitions and applies to any model; ~~tree-specific implementations exploit the model~~ TreeSHAP computes them from the structure ~~for speed. Relative to local-surrogate methods, SHAP is typically denser and, outside tree models, computationally heavier, because many coalition~~

~~evaluations are required.~~of tree ensembles. Anchors searches for a rule that fixes the prediction with a target precision [3]; DiCE searches for a counterfactual instance with the opposite prediction [4].

~~2.3. Evaluating and Comparing Explainers~~

Quantitative evaluation ~~of explanations~~ relies ~~mostly~~ on proxy metrics ~~that need no human judgment: fidelity or:~~ faithfulness under feature removal [~~11–13~~17, 18], robustness ~~of the explanation~~ to small input changes ~~of the input [14], and structural properties such as~~ [19] and sparsity. ~~Toolkits~~, standardized in toolkits and benchmarks ~~standardize these measurements [15, 16], and systematic reviews document how unevenly they are applied [5, 6]. Several results warn against reading a proxy score as correctness: attribution methods~~ [20, 21]. Removal-based scores depend on the removal scheme [22], attributions can pass perturbation tests and fail sanity checks [~~17], removal-based scores depend on the removal scheme [18], and~~ 23], LIME and SHAP can ~~both~~ be manipulated [~~19].~~24], and Shapley ~~value~~ attributions ~~have known limitations~~are hard to read as importance ~~measures~~ when features are dependent [~~20~~25]; feature attribution has provable limits as a basis for inferring model behavior [26].

~~For a fair comparison of LIME and SHAP, confounders unrelated to the explainer must be controlled. This motivates the protocol of Section 3: an identical transformed feature space, shared model artifacts, a fixed class target, and cells matched on (model, seed, N). Under these controls the expected trade-offs become testable: LIME should favor sparsity and speed, and SHAP should favor stability and faithfulness.~~

Recent work compares SHAP and LIME in applied settings, among them education [13], software engineering [14] and mixed-type tabular models [27], and across tabular and image classifiers [6]; their respective strengths are reviewed in [28, 29]. Guidelines and open benchmarks for functionally grounded evaluation on tabular data are emerging [5, 30]. Estimators of Shapley values differ in accuracy and cost [7], LIME's neighborhood sampling can be redesigned to raise its fidelity [16], and counterfactual explainers can disagree on the same instance [31]. This study adds a comparison in which the four explainers are paired on the same instances, their configuration is varied, the validity of the stability measurement is tested, and cost is counted in model evaluations as well as in time.

3. MATERIALS AND METHODS

3.1. ~~Cohort~~Design and Research Questions

All results come from one experimental cohort run for this study under the protocol below (Table 1). An earlier analysis of the SHAP–LIME question used runs collected on several hosts; an audit found that in most of its cells the evaluation instances overlapped the training partition, that one model could not be recovered and that its timings mixed hosts, so no number from it is used here.

For each endpoint Y and each instance, the paired difference is $\Delta_X = Y_{\mathrm{SHAP}} - Y_X$ for $X \in \{\mathrm{LIME}, \mathrm{Anchors}, \mathrm{DiCE}\}$. Three hypotheses state an expected direction; each is tested two-sided, and a difference is read in the stated direction only when its interval excludes zero on that side. Sparsity is reported descriptively:

- **H1 (stability).** SHAP is more stable than X: $\Delta_X(S) > 0$.
- **H2 (cost).** SHAP is more expensive than X: $\Delta_X(C) > 0$, tested in wall-clock time; CPU time and the rows passed to the model are reported alongside, without a test.
- **H3 (faithfulness).** SHAP is more faithful than X: $\Delta_X(F) > 0$ and $\Delta_X(G) > 0$.
- **Sparsity (descriptive).** With its 10-feature budget LIME's sparsity is set by the user, and Anchors returns few features by construction, so a test of sparsity would test a configuration, not a method.

Table 1. The runs analysed here come from the execution cohort released by Herrera-Vásquez and Herrero-Uceda [10], under the protocol and reproducibility contract defined there. The analysis is not the same: that study reports an omnibus comparison across four explainer families at the block level and a paired SHAP–LIME test restricted to 45 cells of three model families. The present analysis covers the 75 matched cells of the five families at the run level, with directional hypotheses, Holm-corrected paired tests, confidence intervals and per-endpoint effect sizes, and it examines how the differences vary with the model family, the dataset and the LIME configuration. Reusing the cohort is deliberate: a fresh execution would add environment variation to a comparison whose purpose is to isolate the explainer.

Four directional questions are addressed, each with its hypothesis, where S is stability, P sparsity, C cost and F fidelity:

- **RQ1 (robustness).** Is SHAP more stable than LIME under input perturbation? $H_1: S_{\mathrm{SHAP}} > S_{\mathrm{LIME}}$.
- **RQ2 (parsimony).** Is LIME sparser than SHAP? $H_2: P_{\mathrm{LIME}} < P_{\mathrm{SHAP}}$, where P is the active-feature ratio (lower is sparser).
- **RQ3 (efficiency).** Is LIME computationally cheaper than SHAP? $H_3: C_{\mathrm{LIME}} < C_{\mathrm{SHAP}}$.
- ***RQ4 (faithfulness).*** *Does SHAP give stronger fidelity and faithfulness signals than*

Design of the benchmark. The primary benchmark is confirmatory; the other parts are exploratory

Factor	Levels
Datasets	Adult (primary, 108 encoded features); Breast Cancer (30), German Credit (61), Diabetes (8), Spambase (57), Phishing (68), Connect-4 (126)
Models	logistic regression, random forest, XGBoost, SVM (RBF), MLP
Explainers	SHAP, LIME (full vector; 10-feature variant), Anchors, DiCE
Seeds	Adult: 5; other datasets: 3
Instances	up to 25 per stratum (TP, TN, FP, FN) of the test split, per model and seed; stability on 3 per stratum
Paired cells	Adult: 25 (model × seed); other datasets: 90
Timing subset	5 instances per stratum, 3 repetitions, one host
Sweep and controls	Section 4.3 and Section 4.4

3.2. Data and Models

Adult [32, 33] is the primary dataset. Six further binary datasets were chosen so that feature dimension crosses feature type: Breast Cancer [34] and German Credit [35] from the UCI repository [36], and Diabetes, Spambase, Phishing and Connect-4 [37–40] in a 2 × 2 of numeric or categorical features and lower or higher dimension. For Connect-4, a win is the positive class and the two other outcomes the negative class.

Each dataset has one fixed stratified split (80/20, seed 42), shared by preprocessing, model training and every seed. Numeric features are scaled and categorical features one-hot encoded, both fitted on the training split only. Training is capped at 20,000 rows (a stratified subsample) so that the SVM fits in reasonable time. The five models [41, 42] are trained once per dataset with fixed hyperparameters (supplementary file) and frozen; every run checks the model file against its recorded hash. A seed therefore changes the evaluation instances and the explainers' randomness, never the model. Evaluation instances are drawn from the test split only, up to 25 from each confusion-matrix stratum; smaller strata are taken whole, and the realized counts are reported. Within a cell, all explainers see the same instances. In small strata, different seeds can draw the same instance; the overlap is measured and reported: the mean Jaccard index of the instance sets of two seeds of the same dataset and model is 0.22 (maximum 0.69).

3.3. Explainers

All explainers receive the same encoded features and explain the predicted~~LIME? $H_4: F_{\text{SHAP}} > F_{\text{LIME}}$.~~

~~3.2. Experimental Design and Analysis Units~~

~~The benchmark has four factors: model family $g \in \{\text{logreg}, \text{rf}, \text{xgb}, \text{svm}, \text{mlp}\}$; explainer $k \in \{\text{LIME}, \text{SHAP}\}$; seed $s \in \{42,123,456,789,999\}$; and sampling intensity $N \in \{50,100,200\}$ instances per stratum of the confusion matrix (TP, TN, FP, FN). This yields $5 \times 2 \times 5 \times 3 = 150$ planned runs, all of which are artifact-qualified (75 LIME, 75 SHAP), forming 75 matched SHAP–LIME cells on identical (g, s, N) coordinates.~~

~~To avoid pseudoreplication, the analysis is hierarchical: metric values are computed per instance, averaged per run, and the paired SHAP–LIME run means on shared (g, s, N) cells are the inferential unit.~~

~~3.3. Data, Models and Explainer Protocol~~

~~All experiments of the main cohort use the UCI Adult dataset [21, 22] with a stratified split and deterministic seeds. The preprocessing pipeline is fitted on the training partition only, with numeric scaling and one-hot encoding of categorical features, which gives 103 transformed features. A fixed set of pre-trained models is shared across all explainer runs to isolate the explainer effect: random forest [23], gradient-boosted trees [24], logistic regression, a support vector machine (SVM) and a multilayer perceptron (MLP). Evaluation instances are sampled by confusion-matrix stratum. Realized instance counts range from 27 to 800 (median 400).~~

~~Both explainers consume the same transformed feature space and target the~~ probability of the positive class:

- **SHAP:** ~~`TreeExplainer`~~TreeSHAP for ~~tree models~~the random forest and ~~`KernelExplainer`~~XGBoost; KernelSHAP for the other models, with ~~a~~50 background ~~sample of $n = 50$~~rows and a budget of 500 coalitions.
- **LIME:** ~~tabular local surrogate~~1,000 samples and the library's default kernel width, $0.75\sqrt{M}$ (≈ 7.79 on Adult). The primary contrast uses the full attribution vector; a variant with ~~a~~ the 10-feature budget shows the effect of ~~`num_`~~the budget.
- **Anchors:** precision target 0.95; the attribution vector marks the features~~=10, an effective `num_samples=1000` and `kernel_width=3`.~~ of the anchor with 1 and the others with 0.

~~These are fixed reference settings, not the library defaults. The sensitivity of the results to the LIME kernel width is examined in Section 4.6.~~

- **DiCE:** one counterfactual of the opposite class (random search); the attribution of feature j is $|x_j - x'_j|$.

Anchors (a rule) and DiCE (a counterfactual) enter as reference explainers of other kinds: the research questions remain about SHAP and LIME, and the two references show whether a SHAP–LIME difference is specific to the pair or shared by explainers in general. These settings are the libraries' defaults where defaults exist, so that a practitioner's first run is what is measured. KernelSHAP's budget is set explicitly because the default ($2M + 2048$ coalitions) made the SVM the cost driver of the whole design; Section 4.3 measures the effect of the default against the explicit budget. Anchors and DiCE return explanations of different kinds, so their vectors measure which features the explanation uses, not signed contributions; the metrics below are comparable across the four methods in that sense.

3.4. Metrics

The metrics follow the protocol of the earlier study [10]. Throughout,Let $f(x)$ isbe the model's predicted probability of the positive class, and $e(x) \in \mathbb{R}^M$ is the attribution vector an explainer returns for instance x with M features, and "; masking" feature j replaces it with its value with the training set mean $\bar{x}_j$. The metrics follow common definitions [17–19]:

1. **Stability** S: robustness of the explanation to small input noise [14]. The explainer is run on $T = 15$ perturbed copies $x^{(a)} = x + \varepsilon^{(a)}$, with $\varepsilon^{(a)} \sim \mathcal{N}(0, \sigma^2 I)$ and $\sigma = 0.1$, and S is the mean cosine similarity over all $T(T-1)/2$ pairs of $T = 15$ explanations of perturbed copies $x + \varepsilon^{(a)}$, $\varepsilon^{(a)} \sim \mathcal{N}(0, \sigma^2 I)$the resulting attribution vectors $e^{(a)} = e(x^{(a)})$,, $\sigma = 0.1$, added to every encoded feature (scaled numeric and one-hot columns alike), with S computed on 3 instances per stratum, Eq. (2):

$$S = \frac{2}{T(T-1)} \sum_{1 \le a < b \le T} \cos\left(e^{(a)}, e^{(b)}\right). \tag{2}$$

- S is 1 when every perturbation yields an attribution vector with the same direction, and near 0 when directions are unrelated.

2. **Sparsity** P: active-feature ratio, the share of the M features whose absolute attribution exceeds the threshold 10^{-4}. A lower active ratio means a sparser explanation.

3. - The cosine with an all-zero vector is set to 0; such vectors are counted and S is also reported without them. Noise on one-hot columns leaves the data manifold; Section 4.4 repeats S with category-preserving perturbations and without noise.

2. **Fidelity** F: a single-feature variant of the faithfulnessthe Pearson correlation of Bhatt et al. [11]. Each feature j, across the M features, between $|e_j(x)|$ and the prediction change $|f(x) - f(\tilde{x}^{(j)})|$ when feature j alone is masked on its own, giving the masked instance $\tilde{x}^{(j)}$, and F is given by, Eq. (3):

$$F = \rho\left(\{|e_j(x)|\}_{j=1}^{M},\ \{|f(x) - f(\tilde{x}^{(j)})|\}_{j=1}^{M}\right),. \tag{3}$$

- where ρ is the Pearson correlation across the M features. F is high when the features an explanation ranks as important are those whose removal moves the prediction most.

4- Absolute values are used because the masked change is unsigned. When either vector is constant, F is undefined and set to 0; these cases are counted.

3. **Faithfulness gap** Δ_k: a deletion test in the spirit of [12, 13]. The G: the prediction change when the $k = 5$ features with the largest $|e_j(x)|$ are masked together, giving $\tilde{x}^{(\text{top-}k)}$, and Δ_k is given by Eq. (4):.

$$\Delta_k = \left|f(x) - f\left(\tilde{x}^{(\text{top-}k)}\right)\right|. \tag{4}$$

- A larger Δ_k means4. **Sparsity** P: the share of features the explanation names first carry more of the prediction.with $|e_j(x)| > 10^{-4}$ (lower is sparser).

5. **Cost** C: per-instance wall-clock explanation time (ms).

LIME's active-feature count is bounded by `num_features=10`, which is a hard ceiling per explanation on one pinned core, CPU time, and the number of non-zero attributions. LIME's sparsity is therefore a consequence of a hyperparameter, not an emergent property of the method. The active-feature ratio measures whether LIME, *as configured*, returns fewer significant features than SHAP; it does not test whether LIME is intrinsically sparser.rows the explainer passes to the model (Section 3.6).

3.5. 3.5. Inferential Protocol

The statistical plan follows established practice for comparing learning methods [25]: a two-sided paired Wilcoxon signed-rank test on the matched cells for each metric [26]; the paired effect size d_z for magnitude [27]; and Holm–Bonferroni control of multiplicity over the five primary metrics [28]. All inferential claims are restricted to artifact-qualified runs: parseable, non-empty results with explicit matched pairing.

Statistical Analysis

Instances are nested in seeds and seeds in models, so instance-level differences are not independent. For each endpoint and contrast, the instance-level differences Δ_{gsi} of model g, seed s and instance i are fitted by a linear mixed model $\Delta_{gsi} = \mu + u_g + v_{gs} + \epsilon_{gsi}$, with random intercepts for the model and for the seed within the model [43]; μ is the effect, reported with its 95% confidence interval and two-sided Wald p-value; if the model cannot be fitted, a t-test on the cell means is used and the case is reported. Because five models are few clusters, the interval is checked by a cluster bootstrap that resamples models [44]. Only instances valid for both explainers of a contrast enter it. On the cell-mean differences, the standardized paired difference d_z [45], the Hodges–Lehmann estimate [46] and the matched-pairs rank-biserial correlation [47], the effect size of the Wilcoxon signed-rank test [48], are reported. The confirmatory family is four endpoints (S, F, G and wall time) times three contrasts; p-values are Holm-corrected over these twelve tests [49]. The four explainers are ranked on the cell means of S, F and G by the Friedman test with the Nemenyi critical difference [50]. The sweep, the controls and the six further datasets are exploratory and are reported without tests.

3.6. ~~Reproducibility~~Cost Measurement and Computing ~~Infrastructure~~Environment

~~The protocol is configuration-driven (YAML files with seeded execution). Each run writes a `results.json` artifact; malformed or empty runs are excluded explicitly and nothing is imputed. All experiments ran on CPU. The reference software environment is Python 3.11 with scikit-learn 1.7.1, XGBoost 3.1.2, SHAP 0.50.0, LIME 0.2.0.1, NumPy 2.2.6, pandas 2.3.3 and SciPy 1.16.3, pinned in the repository. Runs were executed on several commodity workstations under macOS, Windows and Linux, distributed by a work queue. Each run artifact records its configuration, seed, timestamps and duration; processor and memory specifications were not recorded. The consequence for the cost endpoint is discussed in Section 5.3.~~

All runs used one laptop (AMD Ryzen 5 4600H, 6 cores, 31.4 GB RAM, Windows 11), Python 3.11 with scikit-learn 1.7.1, XGBoost 3.1.2, SHAP 0.50.0, LIME 0.2.0.1, alibi 0.9.6 (Anchors) and dice-ml 0.12. Quality was computed in parallel worker processes; quality does not depend on speed. Cost was measured in a separate pass on a subset (Table 1): one process with one thread, pinned to one core, on mains power; the explainers interleaved per instance, after one warm-up explanation each; three repetitions, of which the median is used; and measurements paused while any other process used more than 10% of a core or the machine was in use. Cost is tested in wall-clock time, the time elapsed between the start and the end of an explanation, read from the system's high-resolution performance counter: Windows advances a process's CPU time in steps of 15.6 ms, too coarse for explanations that take a few milliseconds, whereas on one pinned core of a quiet host wall time measures the same work at sub-millisecond resolution. CPU time is reported alongside. A wrapper counts every row the explainer passes to the model. Rows evaluated, unlike time, transfer to other hardware: multiplied by the cost of one model evaluation they give the explainer's cost on any host. For logistic regression and the MLP, whose cost per row is known, this is also given in floating-point operations.

4. RESULTS

4.1. Primary Benchmark on Adult

Table 2 reports the effects on the 25 Adult cells. SHAP is the more faithful explainer: its fidelity exceeds that of each other explainer in all 25 cells (Δ_X from 0.38 to 0.65, $d_z \geq 3.3$), and its faithfulness gap exceeds that of LIME and DiCE (Δ_X = 0.13 and 0.31, intervals above zero). Against Anchors the gap is larger in every cell (0.16), but the interval of the mixed model includes zero (-0.01 to 0.33) because the effect varies between models. On stability SHAP is ahead of DiCE (0.34, in 24 of 25 cells); against LIME (0.18, in 19 of 25 cells) and Anchors (0.10, in 19 of 25) the intervals include zero, so H1 is not supported for these

two contrasts. The Friedman test ranks SHAP first on all three endpoints (mean rank 1.0 on fidelity and on the faithfulness gap, 1.5 on stability; $p < 10^{-8}$ each) and DiCE last (critical difference 0.94); on stability, Anchors (2.0) ranks ahead of LIME (2.8). All-zero vectors occur only for LIME (0.4% of explanations; 5.8% of its perturbed explanations in the stability measure, 6.3% with the full vector); fidelity is undefined for at most 0.1% of the instances of any explainer. Excluding every instance with an all-zero LIME vector, which leaves 134 of the 300 stability instances, changes no conclusion: against LIME the stability difference stays within noise (0.13, 95% CI -0.07 to 0.32), and the fidelity and faithfulness-gap differences keep their size (0.38 and 0.13). The distribution of the instance-level differences is shown in the supplementary file.

Table 2. ~~The analysis covers 150 runs (75 LIME, 75 SHAP) forming 75 paired cells with identical (model, seed, N). Matched coverage spans the five model families (xgb = 15, logreg = 15, rf = 15, mlp = 15, svm = 15) and the three sampling intensities.~~

Primary benchmark (Adult, 25 cells). Per-method mean levels and paired effects Δ_X = SHAP $-X$: mixed-model estimate with 95% CI, Holm-adjusted p, standardized paired difference d_z, Hodges–Lehmann estimate (HL) and matched rank-biserial correlation r_{rb}. Cost: wall-clock ms per explanation

~~4.1. Context: Earlier Results on This Cohort~~

~~The omnibus comparison over this cohort—SHAP, LIME, Anchors [29] and DiCE [30], tested over $n = 15$ blocks per method—is reported in the earlier study [10] and is not reproduced here. It establishes that the four methods are not exchangeable on fidelity or stability, that SHAP leads both endpoints, and that the SHAP–LIME fidelity separation falls marginally below the Nemenyi critical difference. Block aggregation was necessary there because Anchors and DiCE are not complete across the grid (57/75 and 68/75 artifact-qualified runs, respectively, against 75/75 for SHAP and LIME). The same study reports a paired Wilcoxon test of SHAP against LIME on 45 matched cells (logistic regression, random forest and XGBoost), significant on the five metrics in the directions found below. The analysis that follows extends that test to the 75 cells of the five model families. The 45 cells are a subset of the 75, so the two analyses are not independent. The earlier material is context, not evidence, for any claim made in this paper.~~

~~4.2. Primary Paired Inference~~

~~*Table 1 summarizes the paired run-level outcomes on the 75 matched cells. The five primary endpoints are significant under Holm-adjusted testing*~~

~~*Table 1. Matched SHAP-versus-LIME paired results ($n = 75$ cells). For all metrics, Δ = SHAP − LIME, with a 95% confidence interval on the mean paired difference. Sparsity is measured as active-feature ratio, so a positive Δ means*~~

~~*SHAP produces denser explanations. Per-method mean levels for this cohort are reported in [10] and are not restated here*~~

MetricEndpoint	ΔX	SHAP	X	95%Δ_X [95% CI]	p_Holm	d_b L
Stability	LIME	0.74	0.56	+0.718 [−0.01, 0.37]	[+0.663, +0.773]	0.25	2.77 1.00 0.89 1.80
Sparsity (active ratio)	Anchors		0.63	+0.142 [−0.08, 0.28]	[+0.103, +0.181]	0.31	0.40 1.80 4.14 5
Fidelity	+0.248			[+0.236, +0.260]	2.6 × 10⁻¹³		4.82
Faithfulness Gap	DiCE		0.40	+0.045 [0.14, 0.54]	[+ 2.6 × 10⁻¹³ 0.008		0.98 1.36 3
Fidelity	LIME	0.83	0.45	0.38 [0.22, 0.54]	<0.001		0.00 2.40
	Anchors		0.41	0.42 [0.25, 0.58]	<0.001		0.00 1.42 9
	DiCE		0.18	0.65 [0.44, 0.86]	<0.001		0.00 .87
Faithf. gap	LIME	0.40	0.26	0.13 [0.05, 0.21]	0.008		0.00 3.32
	Anchors		0.23	0.16 [−0.01, 0.33]	0.25		0.00 1.96
	DiCE		0.08	0.31 [0.13, 0.50]	0.008		0.00 2.89

Cost ~~(ms)~~	LIME	3,925	170	~~+8047.6~~3,756 [561, 6,950]	~~[+3939.2, +12155.9]~~0.11	~~0.75~~ ~~1.40~~ ~~2.42~~ ~~9~~
	Anchors		9,283	−5,357 [−7,208, −3,507]	<0.001	=1.00 ~~1.88~~ ~~7~~
	DiCE		1,408	2,517 [−961, 5,995]	0.31	~~0~~0.29 ~~1.30~~ ~~4~~ ~~5~~ ~~6~~
Sparsity	all	0.22	LIME 0.98, Anchors 0.06, DiCE 0.02 (descriptive)			

~~The stability effect is the largest separation in the study. The median paired stability difference is +0.866 in favor of SHAP, and the direction is the same in all 75 matched cells. Under the reference configuration, LIME's attribution vectors change direction under small input noise, whereas SHAP's are comparatively invariant.~~

~~For sparsity the sign is also the same in all 75 matched cells. The metric is the active-feature ratio, so SHAP's larger value means denser explanations: LIME is systematically sparser, within the limit stated in Section 3.4.~~

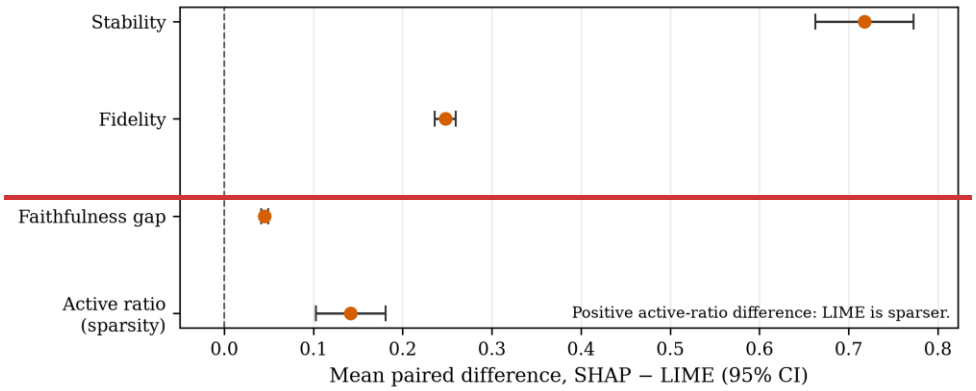
4.2. Cost

Table 3 gives the cost per explanation by model family. Wall time orders the explainers differently by model family. With KernelSHAP, SHAP's median time is 3 to 25 times LIME's: 29.6 against 9.4 ms on logistic regression, 52.2 against 11.0 ms on the MLP, and 19.5 against 0.79 s on the SVM, whose kernel evaluations make it the slowest model for every explainer. These ratios describe medians and are not a test; the test of H2 below pools the five models. With TreeSHAP, SHAP costs about as much as LIME on the random forest (23.0 against 22.3 ms) and a ninth of it on XGBoost (1.4 against 13.3 ms). On the six further datasets, which were timed in the same way (exploratory; supplementary file, S9), TreeSHAP stays 5 to 9 times faster than LIME on XGBoost, but on the random forest its cost depends on the dataset: lower than LIME's on Breast Cancer, Diabetes and German Credit, about equal on Spambase, and higher on Phishing and Connect-4 (23.3 against 18.3 and 49.5 against 28.7 ms). Anchors (0.75 to 5.6 s) and DiCE (1.1 to 2.6 s) are slower than SHAP on every model except the SVM. For H2, the mean difference from LIME is driven by the SVM: the mixed-model interval excludes zero (3.8 s, 95% CI 0.56 to 6.95 s), but the Holm-adjusted p is 0.11 and the Hodges–Lehmann difference is 22 ms, so H2 is not supported against LIME. SHAP is faster than Anchors in all 25 cells, the opposite of H2, and its difference from DiCE is not significant (Table 2). The rows evaluated per explanation, which do not depend on the host, are 1,000 for LIME, 25,001 for KernelSHAP (500 coalitions × 50 background rows, plus the instance), 2,004 for DiCE, 5,501 (logistic regression) to 23,201 (MLP) for Anchors, and none for TreeSHAP. With the 108 encoded features of Adult, a row costs 218 FLOPs for logistic regression and 22,200 for the MLP, so a KernelSHAP

explanation costs 5.5 and 555 MFLOPs, against 0.22 and 22 for LIME. Whether TreeSHAP's cost relative to LIME reverses between XGBoost and the random forest, and whether such a reversal reflects the architecture or only the size of the ensemble, is tested directly: both have 100 trees, and the comparison is repeated with each ensemble at the other's depth (three seeds, 20 instances each; supplementary file, S9). The cost follows the depth of the trees more than the ensemble: lowering the random forest from depth 15 to 6 cuts TreeSHAP's time from 21.7 to 3.2 ms, and raising XGBoost from depth 6 to 15 raises it from 1.4 to 7.4 ms, against 12.5 to 21.6 ms for LIME. At matched depth XGBoost remains 2.3 to 2.9 times cheaper, so the implementation adds a smaller effect. TreeSHAP is thus faster than LIME on both ensembles at depth 6 and on XGBoost at depth 15, and as fast as LIME only on the random forest at depth 15, the setting of the main protocol.

Table 3. Cost per explanation on Adult (one host, one core): median wall time and CPU time, and rows evaluated by the model; theoretical complexity per explanation (T trees, L leaves, D depth; n_s samples or coalitions; n_b background rows). LIME with the full vector; ranges are over the models, the largest always on the SVM; Anchors' rows on XGBoost from a recount (supplementary file, S9). FLOPs for logistic regression and the MLP in the text

~~Fidelity favors SHAP in all 75 matched cells, with a mean paired difference of +0.248 and a very large effect size ($d_z = 4.82$). The faithfulness gap favors SHAP in 74 of 75 matched cells, with a mean paired difference of +0.045 and a large effect size ($d_z = 2.63$). Figure 1 shows the paired mean differences for the four quality-oriented endpoints.~~



~~*Figure 1. Mean paired difference (SHAP − LIME) with its 95% confidence interval for the quality-oriented primary endpoints ($n = 75$ matched cells), as in Table 1. For the active ratio, a positive difference means SHAP is denser, that is, LIME is sparser*~~

~~4.3. Reproducibility across Seeds~~

~~*To characterize how much the readings vary with the seed, the coefficient of variation (CV = SD/mean) was computed across the five seeds of the random-forest stratum at $N = 100$ (Table 2); the medians over all fifteen (N, model) strata are similar (0.9, 1.4, 2.3 and 88.2% in the table's row order). SHAP yields highly reproducible fidelity and stability readings (CV < 1%). LIME fidelity is acceptable (CV = 2.6%), whereas LIME stability has CV = 86.2%. That figure mostly reflects the denominator: when the mean cosine similarity is near zero, any nonzero variation produces an extreme relative deviation. In absolute terms the spread is small (seed SD 0.015): the informative result is the level itself, which is near zero under the reference configuration*~~

Table 2. Coefficient of variation (CV = SD/mean) across the five seeds of the random-forest, N = 100 stratum for the two primary quality endpoints. The CV is uninformative for LIME stability, whose mean is near zero

Method – MetricExplainer	CVModel	Reproducible?Complexity	Wall (ms)	CPU (ms)	Rows evaluated
SHAP – Fidelity(Tree)	RF, XGB	$O(TLD^2)$, no model calls	23.0, 1.4	15.6, 0.8%0	Yes0
SHAP – Stability(Kernel)	0.7%logreg, SVM, MLP	Yes$O(n_s n_b)$ rows	29.6, 19,492, 52.2	31.3, 19,359, 46.9	25,001
LIME – Fidelity	all	$O(n_s)$ rows + fit $O(n_s M^2)$	9.4–786	215.6%–781	Yes1,000
Anchors	all	adaptive (precision target)	749–5,640	742–5,617	5,501–23,201
LIME – DiCEStability	all	iterative search	86.1,064–2%,627	CV not meaningful1,063–2,609	2,004

4.3. Configuration Sensitivity

The sweep varied, on Adult for every model and three seeds, LIME's kernel width and number of samples, and KernelSHAP's budget and background size; KernelSHAP was also run on the two tree ensembles, to separate the method from the TreeSHAP implementation. The grid crosses LIME kernel widths of 0.75, 1.5, 3, 5, 10 and 15 and the default $0.75\sqrt{M}$ with 250 to 5,000 samples, KernelSHAP budgets of 100 to 1,000 coalitions and the default with 10 to 100 background rows, and TreeSHAP with 10 to 100 background rows; each cell has one instance per stratum, with stability on each.

On Adult, LIME's kernel width decides whether its explanations are usable at all. With widths of 0.75 and 1.5 every explanation is all-zero; with 3 and 5 every perturbed copy (15 of 15, and 8.7 of 15 on average) gives an all-zero vector, so stability is not defined. These widths are left out below. With the default width ($0.75\sqrt{M} \approx 7.8$) about one perturbed vector in 15 is all-zero, and with 10 and 15 none. On the remaining configurations all 12 instances of every model are valid, and the comparison uses them (Figure 1).

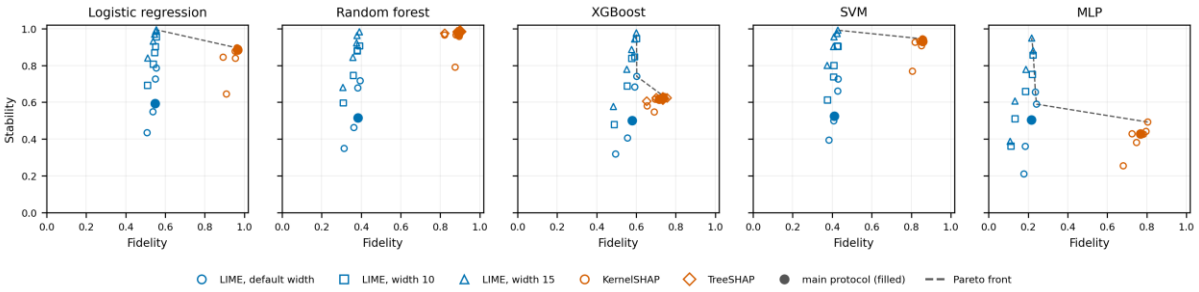


Figure 1. 4.4. Runtime: the Ordering Depends on the Stability against fidelity over the configuration sweep on Adult, one panel per model: means over the 12 instances valid under every configuration; LIME with the full vector, kernel widths 0.75 to 5 omitted as degenerate. Filled markers: main protocol; dashed line: Pareto front

SHAP's fidelity exceeds LIME's in every pair of configurations on all five models, and its faithfulness gap on four; on XGBoost some LIME configurations with a width of 10 or 15 match it (SHAP ahead in 126 of 180 pairs). The stability ordering, in contrast, depends on LIME's configuration. LIME's stability rises with the kernel width and the number of samples, from 0.34 (default width, 250 samples) to 0.98 (width 15, 5,000 samples), while its fidelity changes little (0.36 to 0.44, mostly with the samples). With a width of 15 and 5,000 samples, LIME is more stable than every SHAP configuration on four models; on the random forest TreeSHAP stays as stable (0.98 to 0.99) and dominates every LIME configuration. On the other

models the Pareto front joins SHAP at the fidelity end with wide-kernel LIME at the stability end, so neither explainer dominates.

For KernelSHAP, the library default budget ($2M + 2048 = 2{,}264$ coalitions) and the 500 coalitions of the main protocol give fidelity and stability within 0.01 of each other, except on the MLP (0.81 against 0.77 and 0.49 against 0.43). A budget of 100 coalitions lowers stability by 0.07 to 0.24, and 10 background rows lower fidelity by 0.04 to 0.07. KernelSHAP on the two tree ensembles is within 0.02 of TreeSHAP on both endpoints, so the quality differences between the model families do not come from the TreeSHAP implementation. Cost, timed with the same protocol (supplementary file, S10), depends on LIME's number of samples and not on its kernel width, and on KernelSHAP's coalitions times background rows. LIME's most stable configuration (width 15, 5,000 samples) takes 4.3 to 5.2 times the default 1,000 samples: 47 against 9.3 ms on logistic regression and 3.9 against 0.77 s on the SVM. It then costs more than SHAP on logistic regression (29 ms) and the tree ensembles (TreeSHAP 1.4 and 25.5 ms), about as much on the MLP, and a fifth as much on the SVM (19.2 s). KernelSHAP's library default costs 3.6 to 4.3 times the 500-coalition budget for almost the same quality, and KernelSHAP on the tree ensembles takes 7 (random forest) and 73 (XGBoost) times TreeSHAP's time: the cost difference, unlike the quality difference, does come from the implementation.

4.4. Validity of the Measurements

Five controls test whether the stability and fidelity measurements measure the explainer: (a) explanations of the unperturbed instance repeated T times, which isolates each explainer's own sampling variance; (b) noise that keeps one-hot features valid (a category is resampled instead of adding Gaussian noise to its indicators); (c) rank correlation and top-k overlap on the full vectors, besides the cosine; (d) the effect of excluding all-zero vectors; and (e) the fidelity ordering under four masking schemes (zero, mean, marginal and grouped replacement, where associated features such as education and its numeric code are masked together) on every model. Controls (a) and (b) were run on Adult for every model and three seeds, on the stability instances; (c) to (e) were computed from the stored vectors of the main runs.

Without input noise (a), LIME's repeated explanations are almost identical (stability 0.98 to 0.99 with ten features and 0.89 to 0.97 with the full vector, 0.77 on the MLP), so its own sampling accounts for 8% of its instability under the main protocol (22% with the full vector); the rest is its response to the noise. Own sampling accounts for about two thirds of the instability of Anchors and for all of that of DiCE. With noise that keeps one-hot features valid (b), LIME with the full vector is more stable than SHAP on all five models and in all 15 cells, and LIME with ten features in 13 of 15, whereas under the Gaussian noise of the main protocol SHAP was ahead: on Adult, the stability ordering of SHAP and LIME depends on whether the perturbed rows can occur. SHAP stays more stable than Anchors and DiCE in all 15 cells under both controls. Rank correlation and top-5 overlap (c) keep SHAP ahead of LIME on four of five models, the MLP being the exception. Excluding all-zero vectors (d) raises LIME's cosine stability by 0.01 to 0.08 and changes no ordering. Under all four masking schemes (e), SHAP's faithfulness gap exceeds that of every other explainer on every Adult model, except against LIME on XGBoost, where the difference is close to zero (0.003 to 0.05). On the six further datasets SHAP is ahead in 460 of 480 combinations of model, explainer and scheme; most exceptions are Anchors on Connect-4, which is ahead of SHAP on three of five models under every scheme, as under the main protocol.

4.5. Other Datasets

Table 4 reports the four explainers on the six further datasets, now including SHAP stability. SHAP keeps the highest fidelity on all six, ahead of the next explainer by 0.06 (Breast Cancer) to 0.31 (Diabetes), and the largest faithfulness gap on all six (supplementary file). The stability ordering of Adult does not carry over: LIME with the full vector is more stable than SHAP on every further dataset, by 0.05 (Diabetes) to 0.26 (Spambase), and Anchors is also more stable than SHAP on Phishing and Connect-4, the two

datasets with categorical features in the 2 × 2. Neither the number of features nor their type reverses the orderings of SHAP and LIME: both hold in every cell of the 2 × 2.

Table 4. Other datasets (exploratory): mean over 3 seeds, averaged over the five models; LIME with the full vector. The SD over seeds is below 0.04 in every cell (per cell and per model in the supplementary file)

		Stability	Fidelity						
Dataset		SHAP	LIME	Anch.	DiCE	SHAP	LIME	Anch.	DiCE
Breast Cancer		0.84	0.99	0.38	0.25	0.69	0.50	0.25	0.15
German Credit		0.83	0.94	0.68	0.30	0.77	0.44	0.43	0.18
Diabetes		0.95	1.00	0.82	0.32	0.80	0.49	0.37	0.27
Spambase		0.69	0.95	0.30	0.11	0.64	0.46	0.24	0.13
Phishing		0.75	0.95	0.89	0.25	0.70	0.54	0.51	0.26
Connect-4		0.70	0.85	0.71	0.10	0.80	0.36	0.49	0.09

~~**Model Family**~~

~~The cost difference is significant but strongly heterogeneous. SHAP is slower in 59 of 75 matched cells. The matched-cell median cost is 684.1 ms for SHAP against 65.7 ms for LIME, and the median per-cell SHAP/LIME cost ratio is approximately 5.3 ×. The SHAP distribution is heavy-tailed, so its mean lies far above its median (the per-method means are reported in [10]): the 90th and 95th percentile latencies are approximately 51,456 ms and 67,066 ms, respectively, compared with 12,825 ms and 13,192 ms for LIME. This tail is concentrated in KernelSHAP contexts, especially the SVM.~~

~~The quality ordering (stability, sparsity direction, fidelity, faithfulness gap) is the same in every model family. The cost ordering is not. SHAP is faster in all 15 XGBoost matched cells and in one SVM cell, whereas the logistic regression, random forest and MLP cells are uniformly SHAP-slower (Figure 2, Table 3). The two tree ensembles behave in opposite ways. For XGBoost, `TreeExplainer` was faster than LIME in all 15 matched cells (median 18.1 ms vs. 61.8 ms). For random forests it was slower in all 15 cells (median 1,262.4 ms vs. 335.6 ms). A candidate explanation, not tested here, is that the cost of `TreeExplainer` grows with the number and depth of the trees. For the non-tree models, where SHAP runs in kernel mode, LIME has the lower median runtime (53.3 ms vs. 694.6 ms). "LIME is faster" is therefore a majority statement in this cohort, not a property of the two methods, and "SHAP is fast on tree models" does not hold for tree models in general.~~

~~*Table 3. Median per-instance cost by model group on the matched cells, and the number of cells in which SHAP was the faster method. SHAP uses `TreeExplainer` for the two tree ensembles and `KernelExplainer` for the non-tree models*~~

~~Model group~~	~~Cells~~	SHAP ~~median (ms)~~	LIME ~~median (ms)~~	~~SHAP faster~~
~~XGBoost~~	~~15~~	~~18.1~~	~~61.8~~	~~15 of 15~~
~~Random forest~~	~~15~~	~~1,262.4~~	~~335.6~~	~~0 of 15~~
~~Non-tree (logreg, SVM, MLP)~~	~~45~~	~~694.6~~	~~53.3~~	~~1 of 45~~

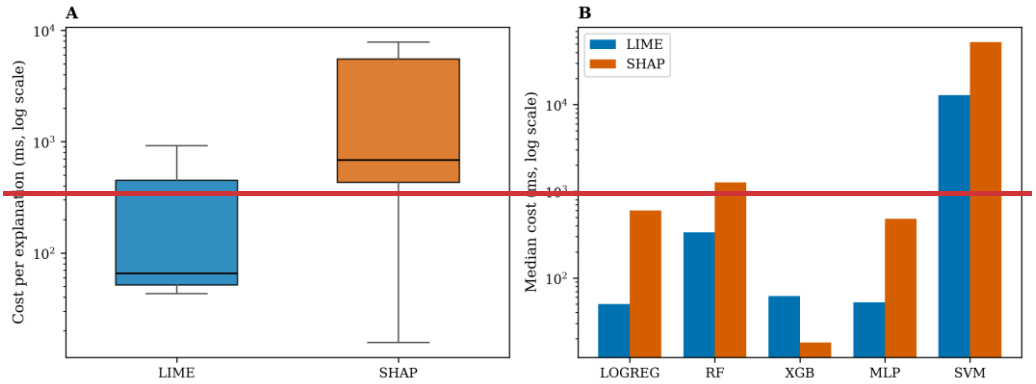


Figure 2. Runtime behavior under matched-cell pairing ($n = 75$). (A) Overall cost distribution by method (log scale). (B) Model-level median cost (log scale), showing the XGBoost exception where SHAP is faster while other families are SHAP-slower

4.5. Cross-Dataset Extension

To probe external validity within tabular classification, a cross-dataset study (EXP3) compared SHAP, LIME and Anchors [29] on two further binary classification datasets: Breast Cancer Wisconsin [31, 32] ($n = 569$, 30 continuous features) and German Credit [31, 33] ($n = 1{,}000$, 20 mixed features). Both used random forest and XGBoost models across three random seeds each, yielding 12 paired dataset–model–seed comparisons per method pair.

Fidelity ordering (SHAP vs. Anchors). SHAP exceeds Anchors in fidelity in all 12 paired combinations (Table 4), with mean gaps ranging from +0.26 (German Credit, XGB) to +0.51 (Breast Cancer, RF).

Table 4. Cross-dataset extension: mean Anchors fidelity (3-seed average) and the paired SHAP − Anchors fidelity gap. The gap is positive in all 12 paired combinations. SHAP fidelity levels for this cohort are reported in [10]; the Anchors levels and the gaps are contributed here

Dataset	Model	Anchors	Gap (SHAP − Anchors)
Breast Cancer	RF	0.265	+0.514
Breast Cancer	XGB	0.208	+0.409
German Credit	RF	0.351	+0.363
German Credit	XGB	0.451	+0.260

LIME on the further datasets. Table 5 reports LIME fidelity and stability on the same datasets, with the paired SHAP−LIME fidelity gap. Two findings follow. First, SHAP > LIME fidelity holds on German Credit (mean gaps +0.23 for RF and +0.24 for XGB, positive for every seed), in the direction found on Adult. On Breast Cancer the fidelity gap narrows substantially: SHAP leads LIME by only +0.07 (RF) and +0.07 (XGB), compared to +0.27 (RF) and +0.21 (XGB) in the corresponding model strata on Adult. Second, LIME *stability* on these datasets (0.75–0.93) is an order of magnitude higher than on Adult, where it is near zero, at the same reference kernel width. The near-zero stability observed on Adult is therefore not a fixed property of LIME.

Table 5. Cross-dataset extension: mean fidelity and stability (3-seed average) for LIME, and the paired SHAP−LIME fidelity gap. Stability is the mean pairwise cosine similarity under $T = 15$ Gaussian perturbations ($\sigma = 0.1$), the protocol

of the main cohort. For comparison, LIME on Adult in the matched model strata: fidelity 0.457 (RF) and 0.553 (XGB); stability ≈ 0.01–0.02 across all model families

Dataset	Model	Fidelity	Gap (SHAP − LIME)	Stability	Sparsity	Cost (ms)
Breast Cancer	RF	0.713	+0.07	0.922	0.331	12.3
Breast Cancer	XGB	0.543	+0.07	0.927	0.331	7.5
German Credit	RF	0.483	+0.23	0.854	0.163	21.1
German Credit	XGB	0.468	+0.24	0.748	0.163	10.2

SHAP stability was not instrumented in the cross-dataset study, so a direct SHAP–LIME stability comparison on Breast Cancer and German Credit is not available.

4.6. LIME Stability Depends on the Kernel Width

A probe on Adult varied the LIME kernel width with everything else fixed (random forest, seed 42, $N = 100$, `num_samples=1000`, `num_features=10`). Table 6 reports the result. Fidelity and stability are computed over the instances that have at least one non-zero weight. The probe covers one model and one seed, so it shows that the dependence exists; it does not measure its size across models.

Table 6. LIME sensitivity to `kernel_width` on Adult (random forest, seed 42, $N = 100$). Reference condition: `kernel_width=3.0`

`kernel_width`	Valid instances	Fidelity	Stability	Sparsity	Cost (ms)
0.25	0/100 (degenerate)	—	—	0.000	30
0.75	0/100 (degenerate)	—	—	0.000	30
3.0 *(ref.)*	96/100	0.518	0.000	0.087	30
10.0	100/100	0.441	0.664	0.093	32

Kernel widths of 0.25 and 0.75 produce zero-weight explanations for all 100 instances, because the exponential neighborhood collapses in the 103-dimensional one-hot-encoded space. The reference width of 3.0 is the smallest of the tested settings that yields usable explanations. Widening the kernel to 10.0 raises stability to 0.664 at a cost in fidelity (0.518 → 0.441). Together with the cross-dataset result, where LIME reaches 0.75–0.93 at the reference width, the probe supports a bounded conclusion: under the reference kernel width, LIME's stability on Adult is near zero; it rises with the kernel width and is high on two lower-dimensional datasets, so it is modulated by the configuration and by the feature space. Adult's one-hot encoding is a candidate cause, and the stability protocol itself may contribute: the Gaussian noise is added to every transformed feature, including the one-hot indicator columns, which then take values that no real instance has. Breast Cancer has no categorical feature. No experiment here separates the encoding from the perturbation scheme, and the two further datasets also differ from Adult in size, class balance and the mix of continuous and categorical features, so this design does not isolate either cause. The near-zero reading should not be taken as a configuration-independent property of LIME.

4.7. Sensitivity of the Fidelity Ordering to the Masking Scheme

The fidelity and faithfulness-gap metrics mask features independently. Adult contains associated features, so a masked instance may fall outside the training distribution, where model outputs are less reliable. To assess whether this changes the method ordering, a probe reused the stored top-10 explanations of the five Adult random-forest runs at $N = 50$, one per seed, and recomputed the prediction-drop metrics under four masking schemes: independent zero masking, mean replacement, marginal empirical replacement, and grouped mean replacement, which masks associated feature sets together (for example `education` and `education_num`). Table 7 reports the result.

Table 7. SHAP–LIME top-k prediction-drop advantage and drop-correlation advantage under four masking schemes (Adult random-forest stratum, five runs at $N = 50$, one per seed). The SHAP direction is positive in all five paired seed cells under every scheme

Masking scheme	Mean SHAP–LIME gap	Mean drop-correlation delta
Independent zero masking	+0.098	+0.515
Mean replacement	+0.098	+0.512
Marginal empirical replacement	+0.081	+0.625
Grouped mean replacement	+0.053	+0.494

SHAP's advantage is positive in all five paired cells under every scheme. Its magnitude, measured as the prediction-drop gap, does not increase, and falls overall, as the masking takes more of the feature dependence into account (+0.098 → +0.098 → +0.081 → +0.053). The drop-correlation delta does not follow this ordering: it is highest under marginal empirical replacement (+0.625) and lowest under grouped mean replacement (+0.494). With five pairs, the exact Wilcoxon test cannot reach $\alpha = 0.05$ under any scheme, so this agreement is descriptive. The probe uses stored top-10 explanations, not full attribution vectors, and is a stress test of the construct, not a replacement for the primary analysis: the direction holds in every pair, and the magnitude depends in part on the masking protocol.

5. DISCUSSION

5.1. What Holds, and Where

Table 5 states each claim with its evidence and its scope. Confirmatory claims rest on the Paired Analysis Addsprimary benchmark and its corrected tests; exploratory claims rest on the sweep, the controls and the further datasets, and are reported as patterns, not as tested effects. The main pattern is that SHAP's advantage lies in faithfulness, not in stability: its fidelity is the highest on every model and dataset tested, and its faithfulness gap on every dataset except Connect-4, where Anchors matches it, whereas the stability ordering depends on the dataset and on the noise. LIME is more stable on all six further datasets, and on Adult as well when the noise keeps categories valid. Cost depends on the SHAP algorithm more than on the explainer: KernelSHAP is the most expensive explainer on the SVM, and TreeSHAP the cheapest on XGBoost. Of the validity controls, only the noise scheme changes an ordering; rank-based stability, the exclusion of all-zero vectors and the masking scheme do not. Over the configuration grid, the fidelity ordering never changes, whereas the stability ordering changes with LIME's kernel width and number of samples: with a width of 15 and 5,000 samples, LIME is the more stable explainer on every model except the random forest.

The four hypotheses are supported on Adult: SHAP is more stable (RQ1) and scores higher on fidelity and faithfulness gap (RQ4); LIME is sparser as configured (RQ2) and cheaper in most cells (RQ3). The analysis on the full grid confirms the direction that the earlier 45-cell test found, extends it to the SVM

and the MLP, and attaches an interval and an effect size to each endpoint; the fidelity and stability differences have the same sign in every cell. SHAP delivers higher stability and fidelity at a substantially higher expected runtime; LIME delivers lower latency and more compact explanations at the cost of robustness.

Two of these results come with conditions, and the conditions are the part most likely to matter in practice. The runtime result (RQ3) reverses between XGBoost and random forests, two models that a guideline would group together as "tree-based". The stability result (RQ1) is measured at one LIME kernel width on one feature space; a wider kernel or a lower-dimensional dataset raises LIME's stability from near zero to between 0.664 and 0.927. A comparison run on one dataset with one configuration would report either of these as a property of the methods.

One candidate reason for SHAP's fidelity advantage is that SHAP ties attribution mass to the model's response to feature removal: local accuracy means the attributions sum to the prediction change relative to the background distribution [2], and removal-based fidelity asks a related question. This mechanism was not tested and carries no guarantee: the masking used here replaces features with reference values and not with SHAP's background expectation, KernelSHAP is itself a sampled estimate, and the advantage is small on Breast Cancer.

5.2. Recommendation Conditioned on the Model Architecture

Table 8 states the recommendation in two tiers. The first-tier criterion is not a preference for a method. It is a function of the model architecture: where a model-aware explainer is also fast, as `TreeExplainer` was for XGBoost, the trade-off between fidelity and latency disappears and SHAP is the better choice in both tiers; where it is not, as for the random forests here, the trade-off remains and has to be measured for the deployed model.

~~*Table 8. Recommendation by deployment tier, conditioned on the model architecture*~~
Table 5. What holds, where, and under which configuration

Claim	**Status**	**Scope**
~~**Tier**~~SHAP more stable than *X*	~~**Recommendation**~~confirmatory	~~**Evidence and condition**~~Against DiCE only, in all five model families of Adult; against LIME and Anchors not significant
SHAP more faithful than *X*	confirmatory	Fidelity against all three in every Adult cell; faithfulness gap against LIME and DiCE, not Anchors
SHAP more expensive than *X*	confirmatory	Not supported: against LIME only with KernelSHAP (3–25 times), not significant after Holm; SHAP faster than Anchors in every cell; DiCE not significant
~~Tier 1: real-time, end-user facing~~Orderings over the configuration grid	~~**SHAP** (`TreeExplainer`) for XGBoost; **LIME** for models without a fast model-aware explainer; for other tree ensembles, measure latency before choosing.~~exploratory	~~For XGBoost, SHAP was faster in all 15 matched cells and more faithful. For random forests it was slower in all 15 cells. For non-tree models LIME has the lower median runtime (Section 4.4). Applies when sub-second latency and compact explanations matter more than maximum fidelity.~~Fidelity: SHAP ahead in every pair of configurations on all five models; faithfulness gap on four (not XGBoost). Stability: depends on LIME's width and samples; LIME (width 15, 5,000 samples) ahead except on the random forest. Widths 0.75 to 5 degenerate
Orderings under the validity controls	exploratory	Stability: LIME ahead of SHAP on Adult with valid one-hot noise; other controls change no ordering. Faithfulness gap: same ordering under all four masking schemes
~~Tier 2: auditing, model analysis~~Orderings on other datasets	~~**SHAP.**~~exploratory	~~Consistent paired advantages in stability, fidelity and faithfulness gap (d_z = 3.00, 4.82, 2.63). Applies to offline or analyst-mediated work that tolerates latencies of seconds to minutes; the sample budget depends on whether SHAP runs in tree-specific or kernel mode.~~SHAP highest on fidelity on all six and on faithfulness gap on five (Anchors equal on Connect-4); LIME (full vector) more stable on all six, Anchors on Phishing and Connect-4
TreeSHAP vs LIME cost by ensemble	exploratory	XGBoost: TreeSHAP faster on all seven datasets. Random forest: faster on three, equal on two, slower on two. Cost follows tree depth: at matched depth XGBoost 2.3–2.9 times cheaper

~~5.3. Validity and Limitations~~

~~**Ranking claims, not validity claims.** The differences reported here are comparative performance within a benchmark.~~ The recommendations are restricted to what was tested. Where faithfulness matters, SHAP is preferable on every model tested: with XGBoost it costs less than LIME and with the random forest about as much (TreeSHAP), and with the other models its median time is 3 to 25 times LIME's (KernelSHAP). Where stability matters more than faithfulness, LIME with the full vector suffices: it is more

stable than SHAP on the six further datasets, and on Adult when the noise keeps categories valid. As feature attributions, Anchors and DiCE are less faithful than SHAP and, except on the SVM, slower. Before relying on a stability comparison, a practitioner should check that the perturbations keep categorical features valid, since Gaussian noise on one-hot indicators reverses the ordering on Adult. LIME's kernel width should be checked next: on Adult, widths up to 5 give all-zero explanations, and the default width gives a stability well below that of a width of 10 or 15 with 2,000 or more samples, at almost the same fidelity. KernelSHAP loses stability below 250 coalitions and fidelity with 10 background rows. The stable LIME setting costs about five times the default, which still leaves it cheaper than KernelSHAP on the SVM but not than TreeSHAP on the tree ensembles.

5.2. Relation to Earlier Work

Earlier comparisons of SHAP and LIME, including a multi-metric benchmark on Adult [51], reported SHAP ahead on fidelity and stability and LIME ahead on cost, for one configuration and with timings from unrecorded hardware. This study differs in four respects: all four explainers are paired on the same instances; every run uses one recorded host and cost is also counted in model evaluations; the configuration is varied and the measurement is tested; and the results extend to six further datasets. The fidelity advantage of SHAP is confirmed. Its stability advantage is qualified: on Adult it is not significant under the original noise and reverses when the noise keeps categories valid, and LIME is the more stable explainer on the six further datasets. The cost statement is qualified as well: LIME is cheaper than KernelSHAP, but not than TreeSHAP on XGBoost. Over the configuration grid, the fidelity statement holds, and the stability statement depends on LIME's kernel width and number of samples as well.

5.3. Limitations

The endpoints compare explainers under a protocol~~. A claim; they do not establish~~ that an explanation is correct~~ for a task would need stronger evidence:~~, which needs ground truth ~~from synthetic or transparent models, perturbation protocols that respect feature dependence, or~~ or human-centered ~~validation [34, 35]. Formal analyses show that several attribution families assign high importance to suppressor variables when features are dependent [36, 37], and faithfulness-style evaluations are informative but do not settle correctness on their own [18, 38].~~

~~**Configuration.** Both methods were run with one fixed reference configuration. The cost gap reflects the method family and the concrete implementation pathway (tree-specific or kernel SHAP). The kernel-width probe~~evaluation [52]. Anchors and DiCE vectors encode which features an explanation uses, not signed contributions. Cost was measured on one host; times scale with the hardware, rows evaluated do not. The study covers ~~one model and one seed. Sensitivity to the SHAP background sample and to the kernel width in other models was not examined.~~

~~**Execution hosts and the cost endpoint.** Cost is per-instance wall-clock time, and the runs were executed on several workstations whose processor and memory specifications were not recorded (Section 3.6). The two members of a paired cell are matched on model, seed and *N*, but did not necessarily run on the same host. Absolute latencies, and the medians reported here, include between-host variance and are indicative of this environment; they are not hardware-normalized measurements. The quality endpoints do not depend on the host.~~

~~**Generalization.** Table 9 states the boundary of each claim. The comparison is limited to~~ tabular binary classification~~; image, text, graph and time-series explanations need different masking schemes and endpoints and were not tested. The endpoints measure how well an explanation tracks the model it explains. They say nothing about whether that model is accurate or fair, and nothing about whether a person decides better with the explanation~~ with five model families and seven datasets.

Table 9. Generalizability boundary of the empirical claims

Claim	Evidence	Generalization status
SHAP > LIME stability	Adult matched cells, one LIME kernel width	Holds for the Adult protocol only; LIME stability is high on two further datasets, where SHAP stability was not measured
SHAP > LIME fidelity	Adult, plus German Credit and Breast Cancer	Supported within tabular classification, with a small gap on Breast Cancer; not established for other data types
LIME lower runtime	Adult, model-dependent	Bounded by the model architecture and the SHAP implementation; `TreeExplainer` reverses the ordering for XGBoost but not for random forests
SHAP > Anchors fidelity	Breast Cancer and German Credit	Fidelity ordering only; not a stability or runtime result
Usefulness for people	No participant study	Not established

6. CONCLUSIONS

On 75 matched cells of the Adult dataset, SHAP shows consistent paired advantages over LIME in stability ($d_z = 3.00$), fidelity ($d_z = 4.82$) and faithfulness gap ($d_z = 2.63$), and LIME produces sparser explanations and runs faster in most model contexts. A cross-dataset extension supports the fidelity direction on two further tabular datasets.

The choice between the two methods is a matter of fit for purpose, and two of the usual rules of thumb need a condition. SHAP is not fast on tree models in general: it was faster than LIME for XGBoost and slower for random forests. LIME is not unstable in general: its near-zero stability on Adult is specific to the reference kernel width and to that feature space. The practical recommendation is to prefer SHAP where stability and fidelity dominate, to prefer LIME where latency dominates and no fast model-aware explainer exists, and to measure latency for the deployed model before relying on either rule.

Future work should instrument SHAP stability on further datasets, vary the SHAP background sample and the LIME kernel width across models, and extend the comparison beyond tabular classification. Whether either methodWhether SHAP outperforms LIME depends on what is asked of the explanation. On the primary benchmark, SHAP is ahead on faithfulness: its fidelity exceeds that of LIME, Anchors and DiCE in every cell of all five model families ($d_z \geq 3.3$), and its faithfulness gap exceeds that of LIME and DiCE; on stability it is ahead only of DiCE (Table 2). The fidelity advantage holds over the whole configuration grid, and the faithfulness-gap advantage on four of five models, whereas the stability ordering depends on LIME's configuration: with a kernel width of 15 and 5,000 samples, LIME is more stable than every SHAP configuration on every model except the random forest (exploratory, Section 4.3). The fidelity advantage also holds on all six further datasets, and the faithfulness-gap advantage on five (Anchors matches SHAP on Connect-4), where LIME is instead the more stable explainer (exploratory, Table 4), as it is on Adult when the stability noise keeps categories valid (Section 4.4). It is paid for in cost only with KernelSHAP, whose median wall time is 3 to 25 times LIME's and 25 times its model evaluations (19.5 s per explanation on the SVM), whereas TreeSHAP is faster than LIME on XGBoost and, on the random forest, faster or slower depending on the dataset, and SHAP is faster than Anchors throughout (Table 3). The difference between the ensembles follows tree depth more than the implementation: at matched depth both ensembles cost less with TreeSHAP than with LIME, except the random forest at depth 15. LIME is therefore the cheaper choice for models that need KernelSHAP, such as the SVM, under a tight latency budget, at the price of fidelity. The faithfulness advantage is not specific to the pair: SHAP holds it against Anchors and DiCE as well, and DiCE ranks last on every quality endpoint. The answer is therefore conditional: the choice of explainer depends on the model architecture, the configuration and the latency budget, and a ranking obtained under one

configuration should not be transferred to another without measuring it. Whether any of these explainers helps people decide is a separate question that needs a study with participants.

Competing Interests

No competing financial interests or personal relationships that could have appeared to influence the work reported in this paper are declared. No external funding was received for this study.

Data and Code Availability

The ~~experimental~~ code, the ~~run artifacts~~frozen models with their hashes, every result row and the analysis ~~exports~~scripts are available in a public repository and ~~in~~ a versioned archive with a DOI, ~~both released~~ under the MIT License; their addresses are withheld for masked review and are given on the title page. ~~The artifacts needed to reproduce this study are the main-cohort results (), the inferential exports (), the cross-dataset results (and), the kernel-width probe () and the masking-sensitivity exports (). Every reported value is re-derived from these files by a verification script that runs in continuous integration~~A supplementary file lists the design, the hyperparameters, the per-dataset and per-model results and the commands that regenerate every table from the result rows.

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